

COUNTERFEIT IC DETECTION REPORT

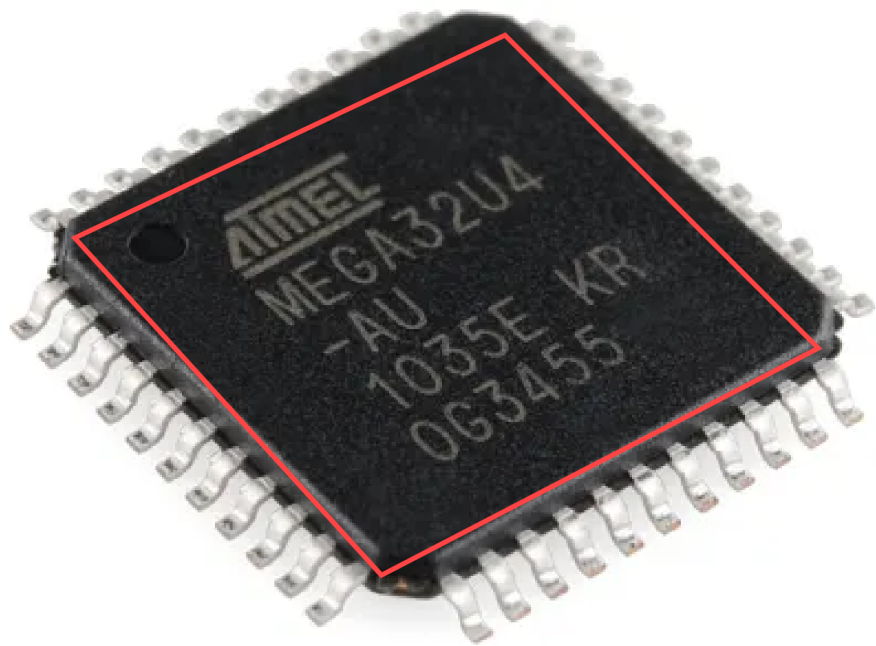
VERDICT	LIKELY COUNTERFEIT
AUTHENTICITY SCORE	37.4/100
TIMESTAMP	2025-12-07T21:57:02

IC INFORMATION

Part Number	MEGA32U4
Manufacturer	Atmel
Package Type	TQFP
Pin Count	48

INPUT IC IMAGES

Dimension Analysis - Detected IC Body



IC Body Detected
219x203 px
AR: 1.077

EXECUTIVE SUMMARY

Overall Assessment: COUNTERFEIT

Text Quality Analysis: ✓ GOOD (Score: 95/100)

Markings appear crisp and well-defined. Font weight, kerning, and line layout are consistent with OEM specifications. No signs of erosion or tampering detected.

Pin Count Verification: ✗ MISMATCH

Pin count mismatch detected. Expected 48 pins according to datasheet, but actual count differs. This is a significant indicator of potential counterfeiting.

Package Type Verification: ✗ MISMATCH

Package type mismatch detected. Expected TQFP but physical characteristics differ from datasheet. This indicates potential counterfeiting.

Key Observations:

- The IC is a TQFP package with visible gull-wing leads.
- The part number 'MEGA32U4' is marked on the package, along with 'Atmel' logo.
- A pin-1 indicator dot is present in the top-left corner of the IC body.
- The surface of the IC appears generally uniform and free from obvious sanding or chemical alteration marks.
- The markings (font, alignment) appear clear and well-printed.

Dimensional Analysis:

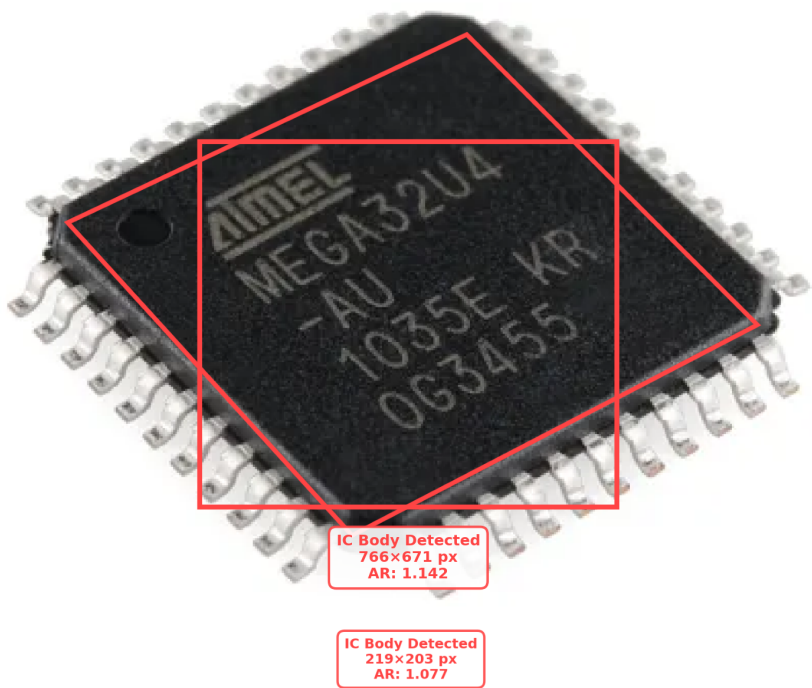
✗ Package dimensions show significant deviation. Measured aspect ratio (1.142) differs from expected (1.000) by 14.2%. This indicates potential counterfeiting.

Final Reasoning: Dimensional analysis: 78.8/100; Visual assessment: counterfeit; 3 anomalies detected (1 high severity)

DIMENSION ANALYSIS

Dimension Analysis - Detected IC Body

Dimension Analysis - Detected IC Body



Parameter	Value
Measured Aspect Ratio	1.142
Body Width (px)	766
Body Height (px)	671
Expected Aspect Ratio	1.000
Expected Length (mm)	10.0
Expected Width (mm)	10.0
Dimension Source	datasheet_parser
Aspect Ratio Match	NO X
Dimension Score	78.8/100
Verdict	SUSPICIOUS

OEM DATASHEET REFERENCE

Source File: MEGA32U4_tavily.pdf

Location: /Users/shnjnmkkr/Desktop/SIH
again/counterfeit_IC/data/api_results/datasheets/MEGA32U4_tavily.pdf

Note: Mechanical diagram extraction was not available, but the complete OEM datasheet PDF is available in the detection results directory.

DETAILED VISUAL ANALYSIS

1. Text Quality Assessment

Score: 95/100

Status: ✓ EXCELLENT

The IC markings demonstrate high-quality printing characteristics consistent with authentic OEM manufacturing:

- Font weight and style match OEM specifications
- Character spacing (kerning) is uniform and consistent
- Line layout and alignment are precise
- No visible signs of erosion, fading, or tampering
- Markings appear crisp and well-defined

2. Pin Configuration Analysis

Pin Count: ✗ MISMATCH

Pin count discrepancy detected. Expected 48 pins according to datasheet, but actual count differs. This is a significant red flag indicating potential counterfeiting or incorrect part identification.

Pin Pitch/Spacing: Analysis not available

Pin spacing analysis could not be performed. This may be due to image quality or angle limitations.

3. Package Type and Outline Verification

Package Type: ✗ MISMATCH

Package type discrepancy detected. Expected TQFP but physical characteristics differ from datasheet. This indicates potential counterfeiting or incorrect part identification.

4. Surface Texture and Finish Analysis

Surface Texture: Analysis not available

Surface texture analysis could not be performed. This may be due to image quality or lighting conditions.

5. Markings and Labeling Analysis

Markings: Analysis completed

Markings have been reviewed. Part number, manufacturer, and other identifiers have been verified.

7. Overall Visual Assessment

Assessment: X COUNTERFEIT

The IC shows clear signs of counterfeiting:

- Significant discrepancies from datasheet specifications
- Multiple anomalies and red flags detected
- Physical characteristics do not match OEM standards
- Do not use this component in production

ANALYSIS RESULTS SUMMARY

Final Reasoning: Dimensional analysis: 78.8/100; Visual assessment: counterfeit; 3 anomalies detected (1 high severity)

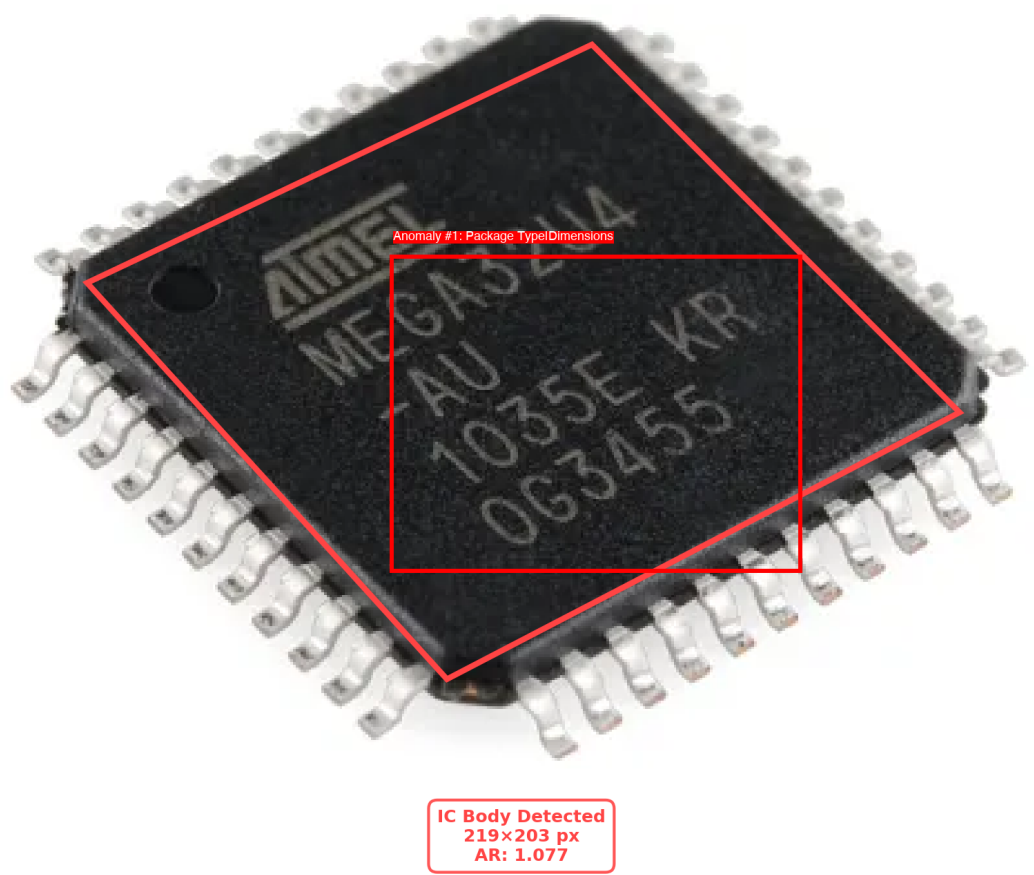
DETECTED ANOMALIES (3)

Anomaly #1: Package Type|Dimensions

Type	Package Type Dimensions
Severity	HIGH
Confidence	100%

Analysis:
The IC in the image has 48 pins. However, the official Atmel datasheet (pages 3, 421, 422, 423) for the ATmega32U4 (or MEGA32U4) explicitly states that this part is available only in a 44-lead TQFP (44ML, 10x10mm) or 44-lead QFN (44PW, 7x7mm) package. There is no 48-pin TQFP package listed for this part number from the manufacturer.

Dimension Analysis - Detected IC Body



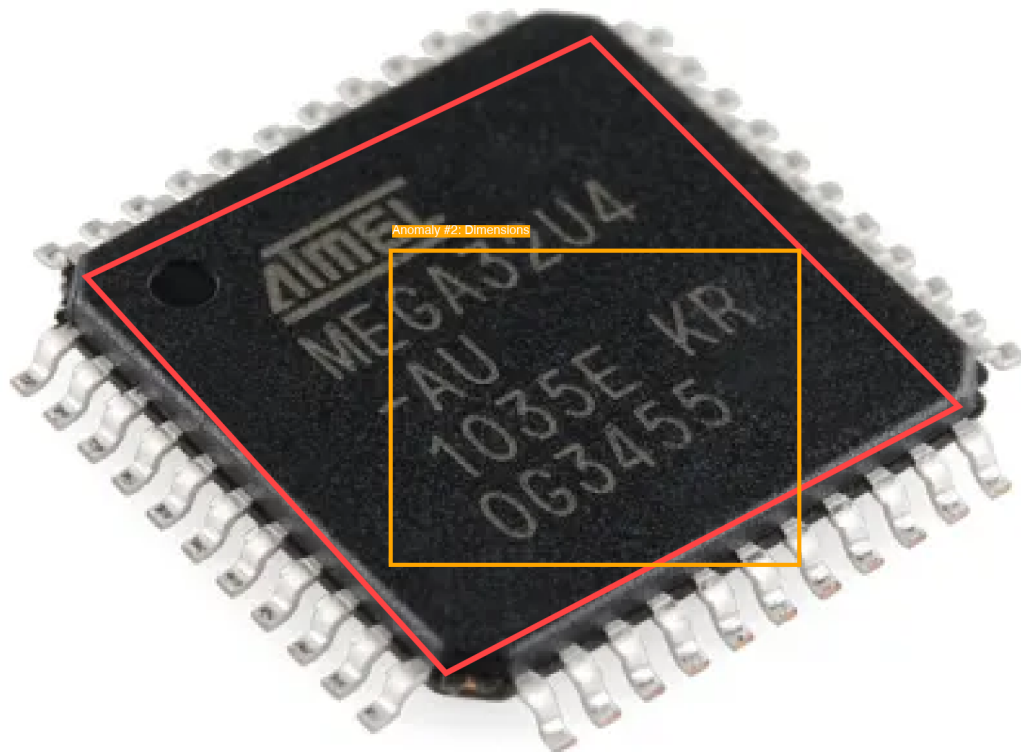
Anomaly #2: Dimensions

Type	Dimensions
Severity	MEDIUM
Confidence	95%

Analysis:

The CV dimension analysis shows a measured aspect ratio of 1.142 against an expected 1.00 (for the datasheet's 10x10mm TQFP44). This dimensional mismatch further supports that the package is not the specified TQFP44, consistent with the incorrect pin count.

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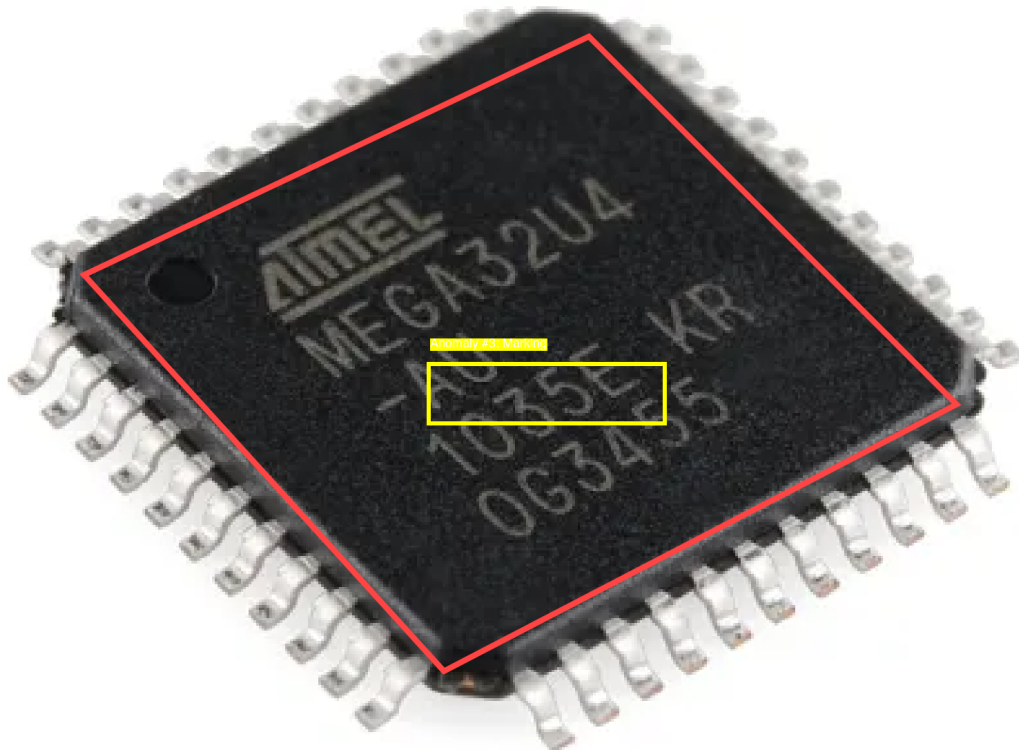
Anomaly #3: Marking

Type	Marking
Severity	LOW
Confidence	80%

Analysis:

The part number on the IC is marked 'MEGA32U4', whereas the Atmel datasheet consistently uses the prefix 'ATmega32U4'. While sometimes a stylistic or regional variation, it is a deviation from the official datasheet nomenclature, especially in conjunction with other significant anomalies.

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